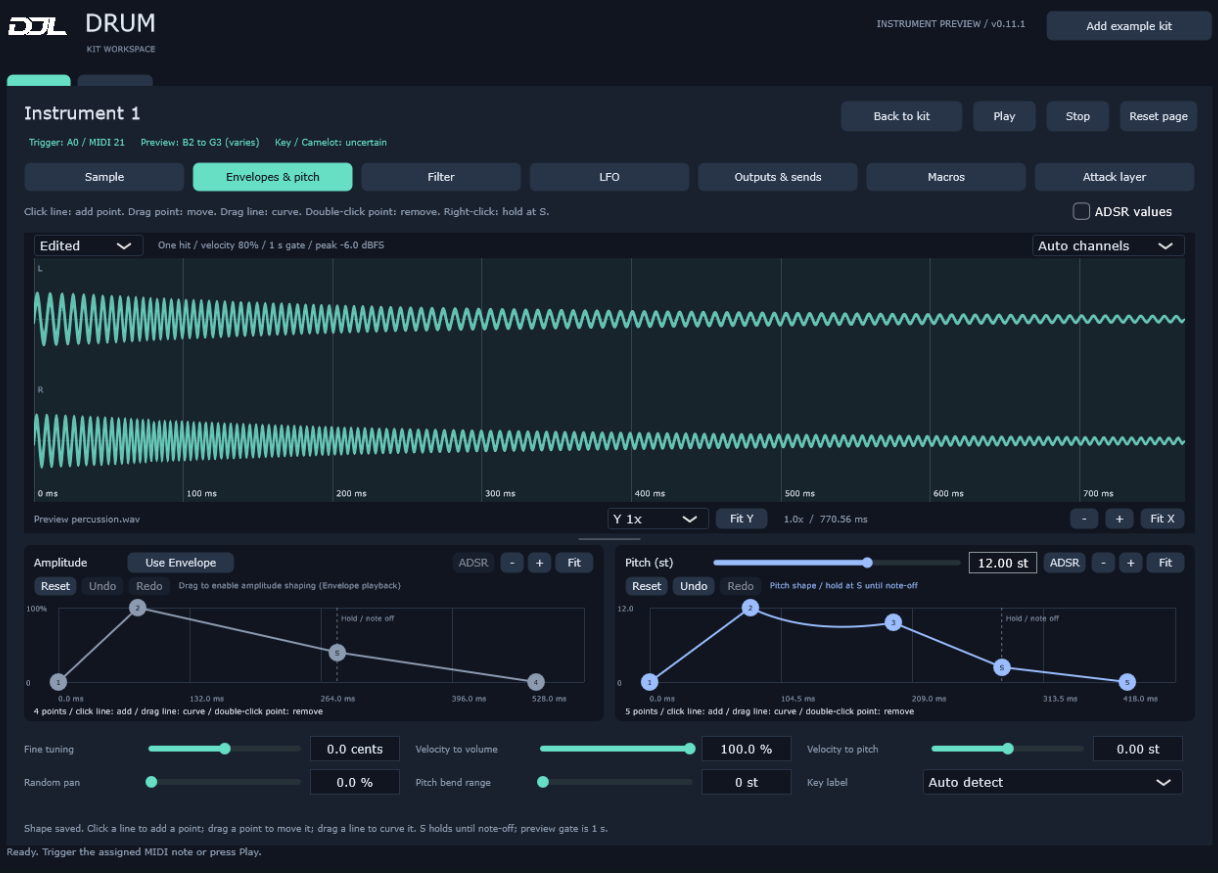
DRUM

The complete user manual

KITS / SOUND EDITING / DAW ROUTING

WINDOWS VST3



Version 0.11.1 editing preview | Instructions, examples and why

DJL Software

29 October 2026

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01 Welcome to Drum

Drum is a sample-based drum instrument for a VST3-compatible DAW. It organises your sounds into named kits, groups and instruments, with a dedicated editor for each sound. This manual covers the Windows 64-bit version 0.11.1 editing preview.

You can keep up to eight kits in one instance. Each kit contains one to sixteen groups and up to 88 instruments in total. One kit plays at a time. Each instrument contains one audio sample and has a unique MIDI trigger note within its kit.

> **Why this matters**
> A kit is a complete set of sounds, a group is a way of keeping related sounds together, and an instrument is one playable hit. For example, a kit called House can contain Kick, Snare and Hats groups. Groups help you find sounds; they do not automatically mix or process audio together.

How to use this guide

Start with Your first playable kit. Then follow the sound-editing chapters in order, or use the clickable contents to find a control. Each chapter explains the actions and the reason to use them. Suggested settings are starting points, not rules for every sample.

Screenshots show the real 0.11.1 interface with demonstration audio and settings. Scroll below the waveform to reach controls that are outside the visible area. The final reference lists all editing values and defaults.

This is a development preview, not complete Ableton Drum Rack or Simpler parity. A separate standalone product is planned. The optional Windows standalone executable is currently a development harness; the VST3 is the main testing target. No factory sample library or finished installer is included.

DJL Software · Manual updated 29 October 2026. Keep the version of this guide with the build it describes.

02 Install, update and load

1. Save your DAW project, then close any application using DJL Drum before replacing the plug-in.

2. Copy the complete DJL Drum.vst3 folder from the supplied package into the VST3 location enabled in your DAW. Preserve its Contents subfolder. The bundle is a folder even though its name ends in .vst3.

3. In Live, open Settings/Preferences > Plug-Ins. Enable the VST3 system folder or your configured VST3 custom folder, as appropriate, and rescan. Keep a single current Drum installation in the locations being scanned.

4. Find DJL Drum under Plug-Ins and load it onto a MIDI track. Open the plug-in editor to see the kit workspace.

> **Why this matters**
> A VST3 instrument turns incoming MIDI notes into audio. It belongs on a MIDI/instrument track. An audio track receiving Drum's extra output is a different part of the routing.

The usual Windows VST3 system location is C:\Program Files\Common Files\VST3. Use your existing dedicated VST3 custom location if that is how this test system is configured. Keep sample libraries and the optional standalone executable outside the plug-in scan folder.

See Ableton: Using VST plug-ins on Windows for host folder and scanning guidance.

Updating an existing project

Keep a backup of important Sets before using a new development build. Existing Drum settings migrate into 0.11.1. New point envelopes and key labels need this version or later; an older build is not expected to understand them. Reopen the Set and confirm sample status, output routing and playback before continuing.

There is no commercial installer, licensing workflow or automatic updater in this preview. If your DAW still shows an older version in the editor header, check for another scanned copy and close/reopen the host.

03 Your first playable kit

1. Open Drum. Rename Kit 1 using the active-kit name box. Press Enter or click away to keep the name; Escape cancels a name edit.

2. Select Group 1 and rename it Drums. Click + Instrument, select its card, then click Load sample and choose a short WAV or AIFF drum hit. Wait for Ready. Alternatively, drop a sample into the empty card area.

3. Click an instrument pad or its assigned piano key to select and play it. Hold the mouse for Gate/Envelope and release to end the held note. Play in the right panel remains a one-second audition; Stop ends sounding and pending hits.

4. Use the Key chooser to assign MIDI 36. Drum calls this C2. In your DAW, draw MIDI note number 36 in a clip, or send that note from a controller.

5. Play the clip. For a connected MIDI keyboard, enable its input in the DAW and arm the Drum MIDI track or set monitoring to In. The matching pad and piano key light while held; short hits flash briefly. A connection alone does not route MIDI into Drum.

6. Add a snare and a hat on different free notes. Save the DAW project once playback works.

> **Why this matters**
> The trigger note selects the instrument; it does not tell Drum what pitch the sample should have. MIDI 36 can trigger any sample. DAWs use different octave labels, so compare the numerical MIDI note when labels disagree.

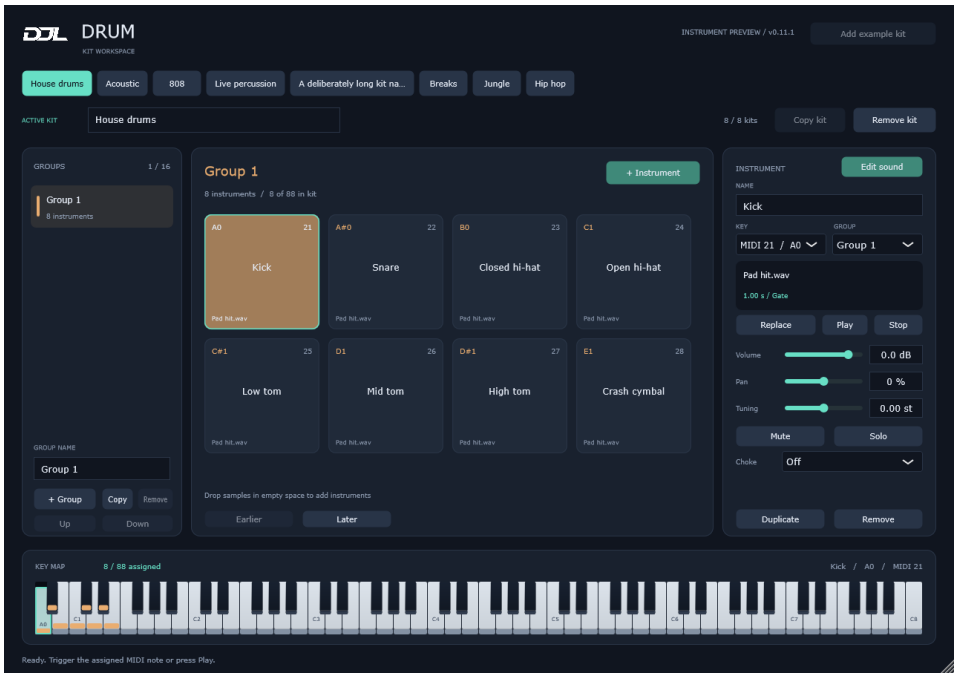
Play auditions at 80% velocity. Gate and Envelope audition use a one-second held note before release. A MIDI clip can use a different velocity or note length, so its result can sound different from the audition.

A useful first comparison

Select the kick, open Edit sound and keep the waveform view on Edited. Change Volume in the kit panel, or Sample gain in the editor, and audition again. The sound and Edited waveform should both respond. Use Back to kit to return to the cards.

If you see a named card with No sample, it is an empty slot. Add example kit creates a layout of named slots; it does not supply recordings.

04 Find your way around



Eight compact named kit tabs, four square pads per row and the playable 88-key piano. Longer kit names use an ellipsis; hover to read the full name.

Area	What to do
Kit tabs	Compact tabs show kit names. Long names shorten with an ellipsis; hover for the full name. Select a tab to activate its kit; + Kit adds one.
Groups	Select a group to show its cards. Rename, copy, move Up/Down or remove it.
Instrument pads	Four across. Press to select and play at 80% velocity; release ends a Gate/Envelope note. Scroll for more pads. Use the right panel to name, assign, load, mix or edit.
Piano	Coloured strips show assignments. Press a key to select/play its instrument. Mouse or incoming MIDI highlights both key and pad; unassigned keys show input but make no sound.
Edit sound	Opens seven tabs with a shared waveform. Scroll the controls area while the waveform stays visible.

Why this matters
The selected pad keeps its outline. Playing adds a bright fill and physically depresses the piano key; release lifts it. Incoming MIDI leaves your editing selection alone. Short hits remain visible briefly. This indicates note input, not audio level: muted or empty slots can still light.

Resize using the bottom-right corner. The minimum editor size is 800 x 600. A larger window makes envelopes and waveforms easier to inspect. Hover over controls for tooltips.

05 Kits, groups and instruments

1. Use + Kit to create another kit, then give it a descriptive name. Click a kit tab to make it active.

2. Use + Group to divide the active kit. For example, separate drums, percussion, cymbals and effects.

3. Select an instrument and use its Group chooser to move it between groups. Earlier/Later changes its order within the group; group Up/Down changes group order.

4. Use Duplicate for an instrument, Copy for a group or Copy kit for a whole kit. The copies keep sound settings and sample references; copied instruments receive free notes where required.

5. Use Remove for unwanted items and read the confirmation. Drum retains at least one kit and one group per kit.

> **Why this matters**
> Use separate kits for alternative drum sets or song sections, and groups to organise the sounds within one set. A group is not a choke group or an audio bus. Renaming or moving an instrument does not change its sample's sound.

Limit	Meaning
8 kits	One active kit at a time; kit switching ends the previous kit's voices.
1-16 groups per kit	Groups can be empty. Their instrument counts need not be equal.
88 instruments per kit	This is the total across its groups, not 88 in each group.
MIDI 21-108	88 unique trigger positions per kit, A0 through C8 in Drum's naming.

New slots and slices take free notes. Assigned notes are unavailable to another instrument in the same kit. Reordering cards does not automatically remap their notes. Switching kits can briefly wait for samples to load; check Ready before playing.

Removing a slot or kit removes its saved configuration, not its original audio file. There is no general undo history inside this preview, so save a project copy before substantial rearrangement.

06 Load and replace samples

1. Select the destination group. Drag one or more actual audio files from Live's Samples browser or Windows Explorer into empty space in the instrument grid to create slots.

2. To replace one sound, drop one file directly onto its card, or select the card and click Replace. An empty slot instead shows Load sample.

3. Wait for the status message before auditioning. Check the filename displayed on the card and in the selected-instrument panel.

4. Rename the slot if a musical role such as Closed hat is more useful than the filename.

> **Why this matters**
> Dropping onto a card replaces that slot's source; dropping into empty grid space creates instruments. This distinction keeps a replacement from accidentally adding a second hit. Replacement preserves the slot's editing controls, so reset any old trim or processing that does not suit the new sample.

Supported audio formats are WAV, AIFF, FLAC, OGG and MP3, with mono or stereo sources. Use audio files rather than Live .adv/.adg devices or presets. Drum does not import Live racks, MIDI clips or arbitrary device chains.

Resource limit	Current behaviour
Per file	Up to 30 seconds and 64 MiB of decoded audio.
Active kit	Up to 512 MiB of unique loaded sample data including waveform caches.
Playback voices	128 simultaneous voices across the active kit.
References	Audio remains in its original file. It is not embedded into the plug-in state.

A failed replacement leaves the previous assignment intact and reports the problem. A missing file on project reload cannot play until relinked. Cloud-only placeholders, disconnected drives or a moved sample folder can make a previously working path unavailable.

07 Save, recall and move a project

1. Save the DAW project to store Drum's kits, names, note assignments, sample paths, edits, custom envelopes, key labels, outputs and macro mappings.

2. Keep the referenced sample files at their saved paths. Before moving a project to another computer, copy the needed audio into a known project sample folder.

3. On the destination computer, load the Set, inspect missing-file messages and use Replace on each affected instrument to choose its copied source. Save the Set again after relinking.

4. Reopen a saved test project once to verify that both Drum and the DAW's receiving/effect tracks recall as expected.

> **Why this matters**
> The DAW saves Drum's settings, but Drum currently saves file references rather than collecting the sample audio. Do not assume a host's Collect All and Save operation has gathered files held privately by this plug-in. Verify the actual copied files before archiving or transferring.

Saved with the instance	Not a portable sample archive
All eight kits, groups and instruments	Original audio file contents are external.
Custom point/curve envelopes and manual key label	Detected pitch/key is recalculated from processed preview audio.
Macro values and mappings; output/send choices	Live effects and receiving-track routing belong to the Live Set.
Sound and mixer controls	Waveform/envelope zoom is an editor view, not a sound parameter.

Edits are non-destructive: trim, reverse, normalise and mono playback do not rewrite the source file. Reset page affects the current page's sound settings. It does not remove macro mappings or undo an earlier file replacement.

08 Volume, pan, tuning, mute and solo

Control	How and why
Volume	Adjust the selected instrument from -60 dB (silent) to +12 dB. Balance the kit here; lower loud hits before raising the whole track.
Pan	Move left/right to place a hit. Stereo material uses balance, not an independent width control. Centre is 0%.
Tuning	-48 to +48 semitones. +12 raises one octave, -12 lowers one. This changes playback speed and duration.
Mute	Silences the instrument. Mute takes priority over Solo.
Solo	Audition one or several instruments in the active kit. Multiple solos are allowed; clear all solos to hear the full kit again.
Reset a slider	Double-click a mixer slider to restore its default. Check the displayed value before playing.

Why this matters
Louder often seems more impressive. When comparing tone changes, roughly match the listening level so you can judge the timbre rather than just the added volume. Summing several samples or adding an attack layer can exceed full scale even when each source file is safe.

Sample gain versus Volume

Sample gain is earlier in the sound path, before filtering and drive. Instrument Volume is later and also scales the post-fader sends. Raising sample gain can change how hard processing is driven; lowering Volume can compensate the final level.

Fine tuning in Envelopes & pitch adds up to +/-100 cents. One semitone is 100 cents. Tune a sustained drum while listening with the bass, rather than expecting every noisy percussion hit to have a stable musical note.

Double-click directly on a slider's numeric value to restore its default, including while the value is being typed. Single-click still opens numeric entry. This also works for Edit Sound, envelope amounts, EQ and macro values/ranges. EQ frequency resets to the selected band's default; macro values reset to 50%, and mapping limits to the selected target's range. Disabled controls remain locked.

09 Choke groups and voice behaviour

1. Select an open hi-hat and choose Choke 1. Select a closed hi-hat and also choose Choke 1.

2. Play the open hat, then trigger the closed hat. The new hit cuts older voices belonging to that choke group with a short ramp.

3. Put another exclusive family, such as alternate cymbal articulations, in a different choke number. Leave unrelated sounds at Off.

4. Use Sample > Retrigger: Cut previous if only repeated hits of the same instrument should cut each other.

> **Why this matters**
> A real closed hi-hat stops the open cymbal's ringing. Choking reproduces that relationship. Organisational groups simply organise cards; choke numbers 1-16 determine which sounding hits exclude one another.

Choke works across the active kit and MIDI channels. Any instrument in the same choke number can cut an earlier member, including an earlier hit of itself. Off allows normal overlap.

Setting	Choose it when
Overlap	You want tails from successive hits to coexist.
Cut previous	You want each new hit of that instrument to replace its prior voice.
Voices 1-128	Limit the overlap available to one instrument; the entire engine still has a shared 128-voice limit.

A large number of long cymbal or loop voices can consume the available pool. Shorten unnecessary tails, reduce per-instrument voices or use suitable choke groups. Stop ends current voices during editing.

10 Waveform views and zoom



The shared waveform and its view, channel and zoom controls. These display controls do not change the audio.

View or control	Meaning
Edited	One simulated hit through current Drum processing, at 80% velocity and a one-second Gate/Envelope note. Updates after edits and macro changes.
Source	Unprocessed audio in the chosen playback direction, with draggable Start/End markers. Reverse flips the picture; Start is the audible beginning. Scaled for visibility.
Send A-F	The chosen send signal before effects on external DAW tracks. An off or silent send can show a flat waveform.
- / + / Fit X	Zoom out/in in time, or show the full duration. Ctrl + wheel zooms around the pointer; ordinary wheel or horizontal scrolling pans left/right.
Y selector / Fit Y	Vertical display magnification, 0.25x-128x, or fit the visible signal. Shift + wheel changes Y zoom. This does not change gain.
Height divider	Drag the divider below the waveform to make it taller; double-click to restore the default height.

Why this matters
At full duration, many audio cycles fit into one screen pixel, so the display shows their extremes. Zoom in horizontally to see individual cycles. Use vertical zoom for quiet sounds instead of making them louder just to see them.

At Edited Y 1x, the amplitude scale is +/-1 (0 dBFS). Keep the same Y zoom while comparing levels. Fit Y can make a quieter sound look equally tall; it is a viewing aid, not normalisation. Zoomed peaks reaching the drawing boundary do not by themselves prove that the audio clips.

Vertical time graticules appear in every waveform view. Labels adapt between milliseconds and seconds as you zoom. Edited timing follows the rendered hit, so pitch and trim can change its duration. The envelope graphs sit below the waveform; each graph has its own envelope-time axis because MIDI determines the held duration.

11 Mono, stereo and what the picture shows

1. Open Sample and choose Source channels: Mono to average the left and right source channels before filtering. Choose Stereo to retain the source's channels.

2. Audition the hit with the rest of the kit, then check the Edited waveform. Use Auto channels to give mono or identical L/R output a single full-height lane.

3. For a stereo source, choose Stereo L / R, Left only or Right only in the waveform display to inspect channel detail. These display choices do not select which channel is played.

> **Why this matters**
> There is no need to make every drum mono. A centred kick or snare can work well as mono, while room recordings, cymbals and stereo effects may benefit from their original width. Decide per sound. Mono averaging can cancel out-of-phase material, so listen as well as looking.

Instrument pan and later stereo modulation still apply after mono conversion. Native Live effects can add stereo space on a receiving track. Source view continues to show the original file even when playback is mono.

Preview boundaries

Edited includes Drum's own eight-band EQ and tone filter, plus gain, trim/reverse, tuning, envelopes, LFO, attack layer, mixer and macros. A filter is a form of frequency shaping: Drum's tone filter offers one modulated response, while its EQ provides eight bands. Effects placed later on Ableton tracks, such as EQ Eight, Compressor or Reverb, are outside Drum and are not drawn in this preview.

The preview is one hit, not a recording of your MIDI performance. Note length, velocity, overlapping voices, pitch wheel and free-running modulation can differ during actual playback. Long renders are capped at 30 seconds or 2.88 million frames and labelled when truncated.

12 Trim, reverse, gain and fades

1. Choose Forward or Reverse on Sample, then choose Source in the waveform view. Source reads left to right in that playback direction. Drag START to the desired audible beginning and END to the desired finish, or enter their numeric percentages.

2. Enable Zero-crossing snap when a trim boundary clicks. It nudges a boundary towards a nearby zero crossing.

3. In Reverse, the waveform flips and START trims from the original file's end. Existing trim boundaries stay on the same source region when you switch direction. Trigger a new hit to hear edits.

4. Adjust Sample gain, Normalise and Fade in/out, then compare in Edited and audition.

Control	Purpose
Start / End	0-100% of the full file in the selected direction. Reverse measures from the original end. Loop percentages follow the direction within the trimmed region. The source file is unchanged.
Sample gain	-24 to +24 dB before processing. Balance source level or feed more/less signal into filter drive.
Normalise	Uses the source peak to scale the sample. It is not loudness matching, compression or a limiter.
Fade in	0-10,000 ms. Softens the sample in Trigger/Gate. Disabled in Envelope, whose amplitude graph shapes the sample. The separate attack layer bypasses both.
Fade out / Gate release	Smooths the physical sample end in all modes and sets Gate release. In Envelope mode, Release controls MIDI note-off; the end fade remains separate to prevent edge clicks.

Why this matters
A tiny fade can remove a discontinuity, while a long fade deliberately changes the instrument's attack or tail. Avoid removing the musical transient when trimming a kick or snare. Check at a useful horizontal zoom and compare before/after by listening.

Trim/loop markers, direction, playback mode and loop on/off apply to newly triggered hits. Many other controls can affect an already sounding voice. For a clear comparison, stop and retrigger after an edit.

The Sample gain box turns red only when the completed processed preview exceeds 0 dBFS. A peak exactly at full scale, a high gain setting by itself, or display magnification does not trigger it. The preview includes envelopes, mixer level and the attack layer, at 80% velocity and a one-second gate. Edited/Send captions use the same limit; Source remains unprocessed. This is one preview hit, not a live meter of the summed kit or external effects. Floating-point values are retained; the warning does not mean Drum hard-clips them.

13 Playback modes and loops

Mode	What happens
Trigger	Plays the hit independently of MIDI note-off. Fade controls shape its amplitude. Useful for one-shot drum programming.
Gate	MIDI note-off begins the release set by Fade out. Useful when note length should shorten a hit.
Envelope	Amplitude follows its ADSR or custom point envelope. Can sustain a loop while held, then release on note-off.

1. For a sustained sound, choose Envelope on Sample and turn Loop On.

2. Set Loop start/end inside the trimmed section. These percentages are relative to the trimmed region, not the entire original file.

3. Raise Loop crossfade a little if the repeated join is abrupt. The effective crossfade is limited to half the loop length.

4. Play a held MIDI note, then release it. Adjust the amplitude release and listen to the tail. Also test short notes.

> **Why this matters**
> Trigger makes a kick's length independent of how long you hold a keyboard key. Gate makes note length useful for articulation. Envelope adds a shaped amplitude contour and looping for sustained sounds. A short non-looping source still ends when its audio runs out.

Envelope is the new name for Classic ADSR; saved sounds keep their mode. It supports drawn shapes and numeric ADSR. Manual Play holds Gate/Envelope for one second. A mouse pad or MIDI note follows your actual press/release. Looping supports both directions; Drum does not time-stretch to song tempo.

Pitch and filter envelopes can act in all three modes, but Trigger ignores note-off. Their release sections need Gate/Envelope note-off to be heard. Amplitude ADSR/custom points affect Envelope only; the graph states when it is inactive.

14 Envelope editing: points and curves



Pitch graph close-up: an extra point and a curved decay segment. The graph has its own amount and time-view controls.

1. Open Envelopes & pitch, or Filter > Filter envelope. Click and release on a connecting line to add a point while preserving its curve. Dragging the line bends it; it does not insert a point.

2. Click and drag a point horizontally for time and vertically for level. Points can move in both directions, within the graph's time/level limits. They can cross neighbours and are reordered in time.

3. Double-click a point to remove it. A minimum of two points is kept; each graph supports up to 32.

4. Press on a line and drag to bend that segment. The bend responds around the place you grabbed; horizontal movement shifts its position and vertical movement adjusts the curve.

5. Right-click a point to mark it S, the level held until note-off. The final point cannot be the hold point.

> **Why this matters**
> Additional points let you shape more than a simple rise and fall: a fast initial spike, a second shoulder, or a slower tail. Curves control how quickly the sound moves between points, even when the points themselves stay at the same times and levels.

The graph's + / - / Fit buttons control its time view. Wheel or horizontal scrolling pans; Ctrl + wheel zooms around the pointer. Middle-button dragging also pans. Fit resumes automatic fitting; the scale stays fixed while dragging. Focus a point and use arrow keys for time/level changes; Shift gives finer steps. Delete/Backspace removes the focused point.

ADSR values reveals the optional numeric stage controls below the waveform. ADSR on a graph discards its custom points and returns to the saved numeric contour. Numeric A/D/S/R sliders are disabled while a custom shape is active. Reset page also clears that page's custom envelopes. Pitch/filter amount remains in the graph header because it sets the depth and direction, independently of the shape.

Reset restores only this envelope's shape, numeric stages and modulation amount to defaults. Undo/Redo keeps up to 64 edits per envelope; an entire drag is one step. View changes are not edits. History lasts while this instrument remains selected in the open editor; changing instrument/sample or recalling different settings clears it. A new edit clears redo. Reset is disabled while a macro owns its timing or amount.

15 Amplitude envelopes and note release

1. Moving, bending or removing an amplitude point automatically selects Envelope playback and Edited view. You can also choose Envelope on Sample or click Use Envelope. Simply adding a point preserves the current sound and mode.

2. For a short drum tail, start with zero Attack, a short Decay, low Sustain and a short Release. Play a MIDI note long enough to hear the decay.

3. For custom editing, move the peak near the start, place S at the desired held level and shape points after S for the release.

4. Test both a held note and a note released during the initial rise. The release starts from the actual current level rather than jumping to the drawn S level.

ADSR stage	Meaning
Attack	Time to rise from zero to full level.
Decay	Time to move from the peak to the sustain level.
Sustain	A level held while the note remains down; it is not a duration.
Release	Time to fall after note-off. A source that ends earlier cannot supply a longer tail.

> **Why this matters**
> Amplitude is the loudness contour. A sharp attack preserves impact; a slower attack softens it. Lower sustain and shorter decay leave more room between hits. Release controls the transition after the key is lifted, not the duration of a Trigger-mode one-shot.

Custom point times run from note-on up to S. After S, point spacing describes time after note-off; the actual held duration comes from MIDI. The graph does not include a fixed-length sustain plateau. The waveform preview uses its own one-second gate, so the two time axes need not align.

Dragging can move points past their neighbours; the graph reorders them in time. Points can also meet to make an immediate transition. Levels stay between 0 and 100%; new point and numeric timing edits stay within the entire original sample duration, irrespective of trim, tuning or MIDI note length. Older longer envelopes keep their sound on opening; their next edit brings points inside the limit. Numeric time edits retain the chosen stage and fit the other time stages into the remaining duration; even a default reset obeys the sample limit. Without a loaded sample the fallback limit is 80 seconds. A nonzero final level fades briefly to zero at the end to avoid leaving an indefinitely active voice.

If a macro controls amplitude Attack, Decay or Release, the graph follows the effective timing and locks point editing. Click Unlink timing in the graph to keep the current timing and remove only this instrument's envelope timing mappings. Other destinations and macro values stay unchanged. The points are then editable. Removing a mapping from the Macros page instead restores its original base value.

16 Pitch envelopes, velocity and expression

1. Open Envelopes & pitch. Choose a positive Pitch envelope amount for an upward offset, or negative for a downward offset. The amount is also visible in the pitch graph header.

2. For a kick pitch drop, start at +12 st, Attack 0 ms, Decay about 40 ms, Sustain 0%, then audition. Increase or shorten the drop to taste.

3. Add or move points for a more detailed contour, or drag a line to change the bend. Audible graph edits switch the waveform to Edited.

4. If the pitch amount is zero, moving/bending/removing points enables +12 st automatically unless a macro controls the amount. Adding a point alone preserves the sound. Set the amount back to zero to bypass the contour.

> **Why this matters**
> A kick often starts higher and settles rapidly into a low body. A pitch envelope recreates that movement. It changes sample playback speed, so a positive offset usually makes cycles closer together and shortens the source. The waveform may change in spacing more obviously than height.

Additional control	What to try
Fine tuning	+/-100 cents for small pitch corrections after coarse Tuning.
Velocity to volume	0-100%; reduce it for more even hits, or retain it for expressive playing.
Velocity to pitch	+/-24 st; stronger MIDI velocities can alter pitch.
Random pan	0-100%; gives hits different stereo positions. Keep centred bass drums controlled.
Pitch bend range	0-24 st; enables the MIDI pitch wheel per channel. Default zero leaves pitch unchanged.

A sample assigned to a different trigger note does not automatically transpose to that note. There is no chromatic key tracking in this preview. Use Tuning, Fine tuning or pitch modulation to change audible pitch. The attack layer has its own Tone control and does not follow sample pitch modulation.

17 Note, key and Camelot indicators



Readout close-up: Trigger, Preview pitch and a separately labelled manual key.

Readout	Interpretation
Trigger	Assigned note name and MIDI number, such as A0 / MIDI 21. This selects the slot.
Preview	Estimated pitch of the processed audio, shown as a note and Hz, or a note range when it varies.
Key (estimate)	An uncertain musical-key estimate from sufficiently tonal preview material, paired with Camelot.
Key (manual)	Your chosen key label. Saved with this instrument; does not tune or otherwise process the sound.
Uncertain / no reliable pitch	The audio does not support a confident result. This is common for hats, snares, noise and short transients.

Why this matters
A single A note does not establish A minor or A major. A drum may have a detectable resonant pitch but insufficient harmony to justify a Camelot code. Keep the key uncertain unless the sample contains useful tonal evidence or you know its key.

1. Read Trigger separately from Preview when tuning. A4 corresponds to 440 Hz in this display.

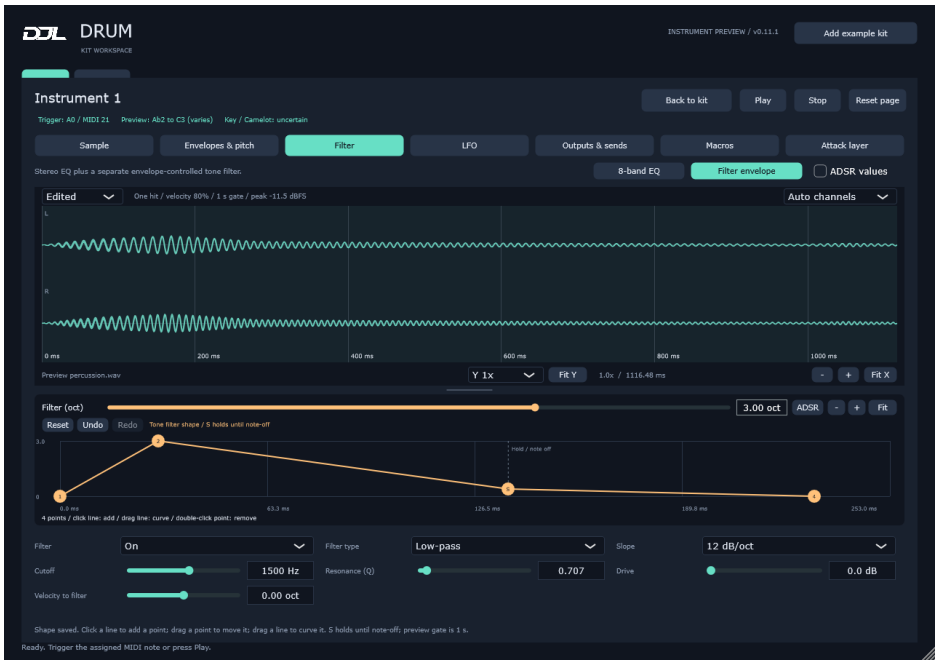
2. If you know the sample's musical key, scroll in Envelopes & pitch to Key label (manual override) and select it. Choose Auto detect to return to analysis.

3. Recheck a manual label after transposing: it remains your entered label and does not automatically transpose with Tuning.

Camelot A codes are minor and B codes are major. For example, A minor is 8A, C major is 8B, and B-flat minor is 3A. The menu may spell an equivalent note with a sharp, such as A# for B-flat. Analysis is an aid, not a guarantee of harmonic compatibility.

Key mapping reference: Mixed In Key: Camelot wheel. Analysis uses a bounded portion of the preview and excludes external DAW effects. In a Send view it analyses that send, which may be silent.

18 Filter and filter envelope



Filter > Filter envelope: the tone filter retains its own cutoff, drive and envelope. It is separate from the eight-band EQ.

1. Open Filter > Filter envelope, then turn Filter On. Choose Low-pass to soften brightness, High-pass to remove low content, Band-pass for a narrower band or Notch to reduce a band.

2. Adjust Cutoff while auditioning. Raise Resonance cautiously to emphasise the area around the cutoff. Try 12 dB/oct or 24 dB/oct slope.

3. Choose a nonzero Filter envelope amount. Use positive octaves to raise cutoff from its base, or negative to lower it. Shape its points and curves using the same gestures as the pitch graph.

4. Use Drive for additional saturation, and compare at a similar final Volume. Use Velocity to filter if harder hits should have a different tone.

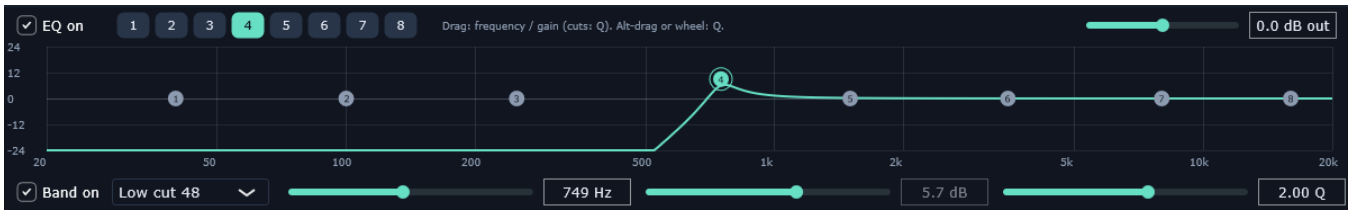
> **Why this matters**
> A moving filter changes brightness through the hit. For example, a positive envelope with fast attack and short decay can let the transient through brightly, then darken the tail. This can create perceived punch without merely raising peak level.

An octave doubles frequency: a +1 octave envelope at full level doubles the base cutoff, subject to the valid frequency range. Amount zero bypasses the envelope movement. Filter Off bypasses the filter and its envelope regardless of the graph shape.

Cutoff is 20 Hz-20 kHz, Q is 0.1-10 and Drive is 0-24 dB. Resonant settings can change level substantially. This is Drum's filter implementation, not an emulation of Live's filter circuit models. An attack layer is mixed after this filter.

Moving or bending a filter point automatically enables the tone filter; if its amount is zero and not macro-controlled, Drum starts at -2 octaves. An audible graph edit selects Edited view. The same click-to-add, line-drag bending, free point movement and double-click removal gestures apply to amplitude, pitch and filter.

19 Eight-band EQ: sculpt the tone



Eight-band EQ close-up. Demonstration low-cut setting; the curve shows the combined stereo response.

1. Open Filter > 8-band EQ. Select band 1-8, choose a response and enable Band on. EQ on bypasses or enables the complete equalizer.

2. Drag a numbered dot horizontally to set frequency and vertically to set gain. Dragging enables that band and the EQ automatically. For cuts and notch, vertical movement sets Q instead of gain.

3. Hold Alt while dragging vertically, or use the mouse wheel over the graph, to change Q for the selected band. Use the numeric Hz, dB and Q boxes for precise values.

4. Double-click a dot to bypass or re-enable its band. Compare the complete result with EQ on, then reduce dB out if boosts leave too little headroom.

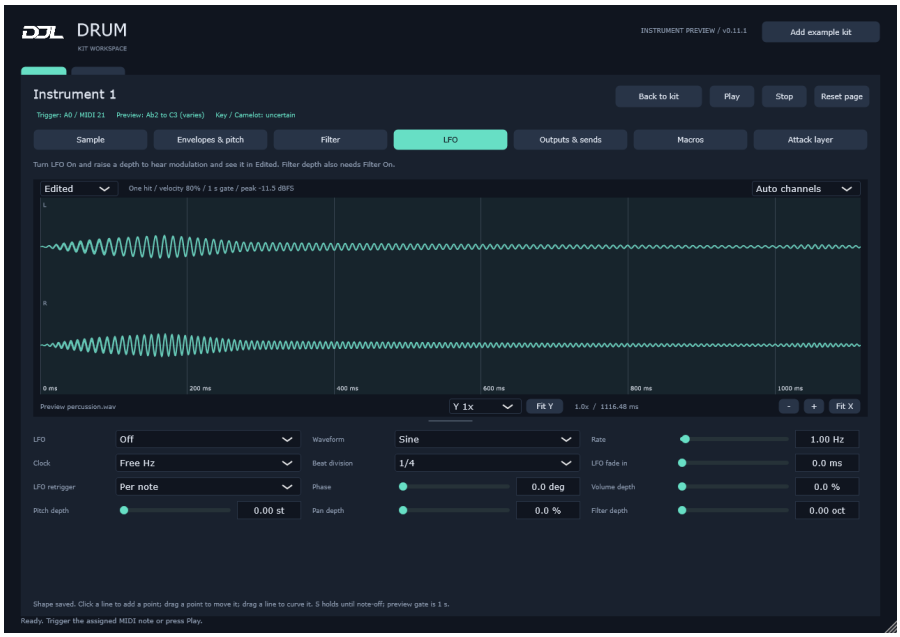
Why this matters
EQ changes which frequencies stand out. Reduce a narrow ringing frequency in a snare, remove low rumble from a hat, or lift the body of a kick. A bell shapes a local region; shelves change a broader low or high range. Lower Q makes a wider, gentler change; higher Q concentrates it. Start with a small cut and compare at similar loudness.

Control	Range and behaviour
Bands	Eight independently enabled bands, sharing one response across left and right. New instruments start with EQ and all bands off.
Responses	Low cut 12/48 dB per octave; low shelf; bell; notch; high shelf; high cut 12/48 dB per octave.
Frequency	20 Hz-20 kHz, limited internally near the host Nyquist frequency.
Gain / Q	Gain -24 to +24 dB for shelves/bell; Q 0.1-18. Cuts and notch have no gain control.
Output	-24 to +24 dB after all bands. A flat/bypassed EQ preserves the original sound.

Try a bell around 300 Hz at -3 dB and Q 1 to reduce a boxy snare. Move the frequency while listening rather than assuming that value fits every sample. The same EQ bands process both the sample and the independent attack layer before their separate envelopes and final mixing. The waveform uses the same EQ processing as playback.

This is Drum's own stereo EQ with EQ Eight-style band editing, not a Live device import or an exact model of EQ Eight. There is no spectrum analyser, L/R or mid/side mode, band audition or adaptive Q in this preview. The envelope on Filter > Filter envelope modulates the separate tone filter, not the eight EQ frequencies.

20 LFO: repeating movement



The LFO page. Turn the LFO on and raise a destination depth; On alone does not add modulation.

1. Turn LFO On. Choose a shape: Sine, Triangle, Square, Saw up, Saw down or Random.

2. Choose Free Hz for a rate in cycles per second, or Host tempo for a musical beat division.

3. Raise one depth at a time: Volume for rhythmic level movement, Pitch for vibrato or warble, Pan for stereo motion, or Filter for changing brightness. Filter depth also requires Filter On.

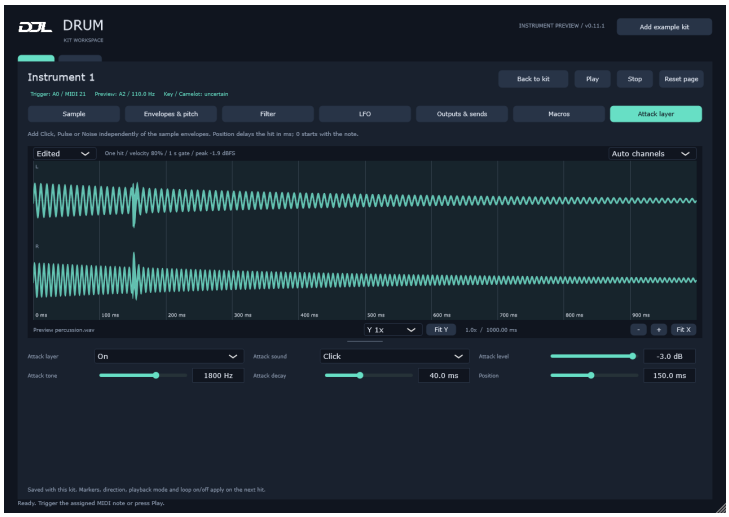
4. Try Per note retrigger for the same motion at each hit. Free-running uses elapsed phase. Adjust Phase to change where the motion begins, and Fade in to introduce it gradually.

> **Why this matters**
> An envelope normally describes one contour after a hit. An LFO repeats. At 1 Hz it completes one cycle per second; a 50 ms drum hit contains only a small part of that cycle, so a slow LFO may make little visible or audible difference.

Free rate is 0.01-30 Hz. Tempo divisions range from 1/64 to four bars. Each voice has its own LFO; Free-running is not locked to the DAW's bar position. Without host tempo, the development harness uses 120 BPM.

For an obvious learning example, use a longer sample, Volume depth around 40%, Sine and 4 Hz. Then reduce the depth to suit the mix. A long fade-in can make modulation inaudible on a short hit. Edited shows its representative audition, not every possible phase of a performance.

21 Add a front click, pulse or noise



Demonstration: a louder, longer click positioned 150 ms after note-on. Use the gentler starting recipe below for a subtle front click.

1. Load a kick sample and open Attack layer. Turn Attack layer On and choose Click.

2. Start with Level -18 dB, Tone 1.8 kHz and Decay 8 ms. Audition, then compare Off/On at a similar overall level.

3. Raise Tone for a brighter tick or lower it for more knock. Adjust Decay between 1 and 100 ms to change the transient's length.

4. Set Position to 0 ms for the front of the note. Try 5-15 ms for a slightly later knock, or 50-150 ms for an obvious second hit. The range is 0-2,000 ms. Trigger a new note after changing Position.

5. Try Pulse for a short sine burst, or Noise for a rougher front edge. Zoom Edited into the first few milliseconds to see the addition.

> **Why this matters**
> The low body of a kick can be strong yet difficult to hear on small speakers or through a dense mix. A short upper-frequency transient can make its start easier to perceive without extending the bass tail.

Click adds a harmonic, Pulse is tonal, and Noise is a repeatable filtered burst. Attack level now sets the burst peak: 0 dB is full scale before EQ, velocity, LFO and mixer; -6 dB is roughly half amplitude. -60 dB is silent. Tone is 30 Hz-16 kHz, limited by sample rate.

The layer has its own short decay window and bypasses the sample amplitude, pitch and filter envelopes, fades and trim. It fires once per note at Position, even if the sample or its amplitude envelope has already ended. Normal note-off does not cancel it. Stop, mute, choke, retrigger and voice stealing can still cut it. It requires a loaded sample and does not repeat at loop boundaries.

Tone sets pitch independently. EQ, velocity, LFO volume and the mixer still affect the result: Play/Edited use 80% velocity, so an isolated 0 dB attack normally previews at -1.9 dBFS. Adding the sample can exceed full scale. Edited/Send show the layer; Source does not. Position is saved per instrument and defaults to 0 ms in older projects.

22 Slice a sample into instruments

1. Load the audio you want to divide. Set its Start/End range on Sample.

2. Scroll to the slicing controls. Choose 2, 4, 8, 16, 32 or 64 equal slices, then click Slice to new group.

3. Return to the kit workspace and select the new slices group. Audition the new instruments and inspect their free MIDI assignments.

4. Rename useful slices and refine their Start/End points individually. Save your project when the mapping is ready.

> **Why this matters**
> Slicing lets different MIDI notes trigger different regions of one recording. This can rearrange a loop into a new rhythm while keeping the original file and instrument available.

Slices divide the selected region equally in time. This preview does not detect transients or warp slices to the host tempo. A slice boundary may fall between beats; use a sensible trimmed range and inspect each hit.

The operation creates a new group and keeps the source instrument. It needs enough of the kit's 88 slots and a free group within the limit of sixteen. If there is insufficient room, the operation is rejected as a whole.

Each slice shares the source file and inherits sound controls, including custom envelopes, outputs and macros, but starts in Trigger with looping off. A copied amplitude envelope is therefore inactive until you choose Envelope. Copies reference the audio rather than creating new audio files.

23 Outputs, sends and native Live effects

1. For a kick needing its own Live effect chain, open Outputs & sends and choose Out 2. Leave other instruments on Main or assign other outputs.

2. In Live, create an audio track. In Audio From, choose the track containing Drum, then choose the corresponding Drum output in the lower chooser. Set Monitor to In and route that track to Main.

3. Put native Live audio effects, such as EQ Eight or Compressor, on that receiving audio track. Use Out 3 and another receiving track for a separate snare chain.

4. For a shared effect, raise Send A on several instruments and receive Drum's Send A on another audio track. Add a wet reverb or delay there; keep the direct instrument outputs for the dry signal.

> **Why this matters**
> Use a dedicated output for a private effect chain: Kick on Out 2 can have Compressor, while Snare on Out 3 can have EQ Eight and Saturator. Sounds sharing an output share its chain. A send adds a parallel copy for shared effects such as room reverb. Drum supplies the audio; Live owns the devices.

Output facility	Current behaviour
Main + Out 2-16	Sixteen main/aux destinations. Main can be mono or stereo; auxiliary buses are stereo.
Send A-F	Six additional stereo buses with per-instrument post-fader send levels. -60 dB is Off.
Disabled destination	A disabled instrument output falls back to Main; a disabled send is silent.
Enabled but unreceived	An instrument assigned to an enabled auxiliary leaves Main even if you have not connected a receiving track.

Live owns native devices, so they cannot be dropped into Drum. Third-party effects can also run on the receiving Live tracks; this build does not host VST3 effects internally. There are sixteen instrument-output destinations, not 88 private chains. Organisational groups do not create buses.

Host routing reference: Ableton: Routing and I/O. Output names may include channel numbers. External effect audio is not included in Drum's waveform; monitor the receiving track to hear it.

24 Macros and automation

1. Open Macros for the selected instrument. Choose a Target for Macro 1, then set Min and Max to a useful range.

2. Move Macro 1's value and audition. For example, map Filter cutoff from 300 Hz to 5,000 Hz with the filter enabled.

3. Select another instrument and map its Macro 1 to a different target or range if one gesture should change both sounds.

4. Expose Macro 1-16 in the DAW's plug-in parameter controls and record or draw host automation. Test playback of that automation.

> **Why this matters**
> A macro is a single control that can coordinate changes across a kit. The mapping belongs to each instrument, while the sixteen macro values belong to the plug-in instance. A new kit can interpret the same macro value differently.

Each instrument has one target per macro. Reversed Min/Max ranges are allowed. If several macros map to the same target, the highest-numbered mapping wins. Unmapped macros do nothing.

Target families	Examples
Mixer	Volume, Pan, Tuning.
Sound	Sample gain, Filter cutoff, Resonance.
Envelopes	Amplitude Attack/Decay/Release; Pitch and Filter amounts.
LFO	Rate, Pitch depth and Volume depth.
Sends	Send A through Send F.

A mapped target uses the macro's effective value instead of its stored base control. Removing the mapping restores the base value. Reset page does not clear mappings. Custom amplitude timing mappings stretch the relevant curve sections; the graph follows that timing and locks direct point editing while a timing mapping is present. Unlink timing on the envelope graph instead keeps the currently heard timing while releasing only its timing mappings.

The sixteen stable macro parameters support host automation. Direct automation lanes for every per-instrument field, macro names and macro variations are not supplied in this preview.

25 Practical sound recipes

A kick with more thwack

1. Keep the initial sample transient when trimming. Try Mono if the stereo source loses focus, then compare in context.

2. Add Click at -18 dB, 1.8 kHz, 8 ms and Position 0 ms. Increase only until it is clear; try Position 5 ms to separate the click slightly from the sample.

3. Try a +12 st pitch envelope with zero attack, around 40 ms decay and zero sustain. Use a curved decay if the pitch sweep sounds too obvious.

4. Balance final Volume and compare with bass and hats playing.

> **Why this matters**
> The click and pitch drop do different jobs: the click adds high-frequency definition, while the pitch contour changes the kick body's movement. Too much of either can turn a natural kick into a conspicuous effect.

Shorten a ringing snare

Choose Envelope, keep the amplitude attack fast, lower Sustain and shorten Decay. Test with a long MIDI note so note-off does not mask the decay setting. Then set Release to avoid an abrupt cutoff on short notes. If the original sample is already short, lengthening an envelope cannot recover missing audio.

Natural hi-hat interplay

Put open and closed hats in the same choke number. Leave kick and snare outside that choke group. Use different velocities for rhythmic accents; avoid excessive Random pan if the pattern needs a stable stereo position.

A shared room

Send several instruments lightly to the same external reverb bus. Use a mostly wet effect on the receiving track to avoid doubling the dry path. Keep low-frequency drums drier if the room tail clouds the rhythm.

A swelling reversed cymbal

Reverse the sample, trim to a useful length, then adjust Fade in and end fading. Use a MIDI clip to place the swell's arrival precisely. Tuning changes its length; there is no warp-based duration lock.

26 Troubleshooting sound and display

Symptom	Check and action
No sound	Wait for Ready; verify a loaded file, active kit, assigned MIDI number, input monitoring, Mute/Solo, Volume, sample gain and output routing. Try Play.
Play works but MIDI does not	Compare MIDI numbers rather than octave names. Check the DAW track's input, arm/monitor state and clip notes.
Pads light but no sound	Highlights show note input even for muted/empty slots. Check the loaded sample, active kit, mute/solo, output routing and track monitoring.
A hit disappears after routing	Connect the chosen auxiliary output to a receiving track. If the host enables it, audio may already have left Main.
Missing sample after reopening	Use Replace to relink the external file, then save. Check drive/cloud availability.
Pitch graph changes but sound does not	Check Pitch amount, macro overrides and the preview view. Numeric ADSR at zero amount is silent; audible point edits enable +12 st unless mapped.
Amplitude graph is grey	Choose Envelope or Use Envelope. Trigger/Gate use fade controls.
Filter/LFO makes no difference	Enable the processor and raise its amount/depth. Filter modulation also needs Filter On. A slow motion may be subtle on a short hit.
Waveform seems unchanged	Use Edited. Zoom in in time for pitch or filter detail. External Live effects are not part of the preview.
Waveform is too small	Try Fit Y, Y zoom, a taller divider position, or Left/Right only for full-height channel detail.
Sound clips or crackles	Lower summed levels/drive/attack layer; inspect trim and loop joins. If dropouts affect the whole project, check the DAW driver/buffer and system load.

After changing a latched playback setting, stop and play a new hit. Compare with a simple unprocessed sample to distinguish a source problem from a routing or modulation problem.

27 Troubleshooting editing and recall

Symptom	Reason and next step
Moving points changes their numbers	Points can cross neighbours and are reordered by time. S follows its held point unless moved to the final position, where the preceding point becomes S.
Line adds a point instead of curving	Drag a line to bend it, with no minimum hold time. Click and release without dragging to insert a point. Drag an existing point to move it.
Point or stage will not move further in time	The original sample duration is the editing limit, even if trim or tuning changes playback length. MIDI hold time is separate.
Cannot add/remove more points	Each graph keeps at least two and at most 32 points.
Numeric ADSR is disabled	A custom shape is active. ADSR restores the saved numeric contour and removes its custom points.
Graph is locked	A macro controls its timing. Click Unlink timing on the graph to retain the current timing and release those assignments, then drag points or lines.
Control reset has no audible effect	A macro may still override it. Also check the correct kit/instrument and the mode/processor enable setting.
Note or key is uncertain	Noise and short/changing transients may lack stable pitch; a single tone does not establish major/minor. Use a known manual key only when appropriate.
Manual key is wrong after tuning	Manual labels are metadata and do not transpose automatically. Update the label yourself.
Copy/slice cannot complete	Check 8-kit, 16-group and 88-instrument capacity and available notes.
Changes absent in a reopened Set	Save the DAW project after editing, confirm the intended instance/build and sample paths, then verify by reopening.

Reporting a reproducible issue

Record the Drum version shown in the header, Windows and DAW versions, audio sample rate/buffer, the selected playback mode, output routing and exact steps. Include whether the problem happens with Play, MIDI or both. A screenshot and a short sample you are allowed to share are useful. Keep a project backup before simplifying it for investigation.

This preview has automated processor/state/preview checks and compiled VST3 loading checks. These do not guarantee every host, hardware or controller workflow. Review the new envelope gestures, output routing and project recall in your own Live Set.

28 Shortcuts and editing reference

Action	Control
Rename	Enter or click away commits; Escape cancels the text edit.
Mixer default	Double-click Volume, Pan or Tuning slider.
Waveform time zoom	Ctrl + wheel; or - / + / Fit X.
Waveform time scroll	Wheel or horizontal scrolling.
Waveform vertical zoom	Shift + wheel; or Y selector / Fit Y.
Waveform height	Drag divider; double-click divider restores default.
Add envelope point	Click and release on a segment.
Move envelope point	Drag in two dimensions; points can cross neighbours and are reordered.
Remove envelope point	Double-click, or focus it and press Delete/Backspace.
Bend a segment	Drag the line; horizontal position influences bend placement.
Set held level	Right-click a non-final point to mark S.
Precise point adjustment	Left/Right: 1 ms; Up/Down: 1% of level. Shift: 0.1 ms / 0.1%.
Envelope time view	- / + / Fit; Ctrl + wheel zooms; wheel or middle-drag pans.
Return to ordinary ADSR	ADSR on the graph.
Envelope reset / undo / redo	Reset / Undo / Redo on each graph. One drag is one edit.
Reset current editing page	Reset page. Does not remove macro assignments.
Audition / stop	Play / Stop. Play uses 80% velocity and a one-second gate where applicable.

Zoom controls do not change audio. Point/envelope controls do. Changing a point from a keyboard follows the displayed direction for negative modulation amounts.

29 Signal path and current limits

The sample region passes through pitch/loop playback, source gain/normalisation, Mono/Stereo, tone filter/drive and eight-band EQ, then its amplitude envelope/fades. The attack layer uses a separate clock, its own EQ state with the same band settings and its own decay window. The two signals then combine before shared velocity response, LFO volume and the instrument mixer. Post-fader sends feed their separate buses.

Why this matters
Raising Sample gain can push Drive harder; decreasing final Volume turns the processed result down. The attack layer keeps its click even when the sample has a long fade or a quiet envelope. The eight-band EQ affects both paths. To soften the layer itself, adjust its Level, Tone or Decay.

Facility	Scope of version 0.11.1
Sources	One mono/stereo audio sample per instrument; no velocity-layer stack or nested instrument devices.
Timing and pitch	Resampling changes pitch and duration together. No Warp/time-stretch or chromatic key tracking.
Slicing	Equal regions only; no transient, beat or manual slice-marker engine.
Envelopes	Amplitude, pitch and filter; ADSR or 2-32 custom points, curves and hold point. No repeating envelope loop mode.
Effects	Tone filter/drive, eight-band stereo EQ and attack generation inside Drum; native DAW effects via separate outputs. No internal return reverb/delay or plug-in hosting.
Automation	Sixteen macros; no full per-field automation matrix or macro variations.
Platform	Windows x64 VST3 preview. macOS build scaffolding needs native validation; standalone product is planned.
State and samples	Settings saved by the host; external audio files must be retained/relinked.

Groups are organisational, not layered device chains. MIDI notes 21-108 select unique instruments. The design is a clearer kit workspace with sound shaping; do not assume every function of another drum rack is present.

30 Glossary and further reading

Term	Plain-language meaning
Sample	A recorded audio file played by an instrument slot.
MIDI note / velocity	A trigger message and its strength value. MIDI contains instructions, not recorded sound.
Voice	One currently playing occurrence of a sample. Repeated hits may overlap as separate voices.
Transient	The short initial change at the start of a hit.
Envelope / sustain	A one-hit contour; sustain is the held level, not a time.
LFO	A repeating control signal that moves a sound parameter.
Semitone / cent / octave	Pitch distances: 100 cents per semitone; twelve semitones per octave; an octave doubles frequency.
Hz / ms	Cycles per second / thousandths of a second.
dB / dBFS	A level ratio / level relative to digital full scale. 0 dBFS is the reference peak limit for ordinary fixed-point audio.
Cutoff / Q	Filter transition frequency / resonance strength or selectivity.
Choke group	A set of instruments whose new hits stop older members.
Send / post-fader	A parallel output feed / one whose level follows the instrument's final volume.
Camelot	A musical-key code: A for minor, B for major. It does not identify a MIDI octave.
Non-destructive	Processing preserves the original audio file.

For host routing, see Ableton Routing and I/O. For installation, see Ableton VST on Windows. For the key chart, see Mixed In Key Camelot wheel. Third-party names identify their respective products or owners.

31 Control reference: Sample

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
Start	0 to 100 %	0 %
End	0 to 100 %	100 %
Sample gain	-24 to 24 dB	0 dB
Source channels	Stereo, Mono	Stereo
Direction	Forward, Reverse	Forward
Normalise	Off, On	Off
Zero-crossing snap	Off, On	Off
Playback	Trigger, Gate, Envelope	Trigger
Voices	1 to 128	128
Retrigger	Overlap, Cut previous	Overlap
Loop (Envelope)	Off, On	Off
Loop start	0 to 100 %	0 %
Loop end	0 to 100 %	100 %
Loop crossfade	0 to 500 ms	0 ms
Fade in	0 to 10000 ms	0 ms
Fade out / Gate release	0 to 10000 ms	2 ms

32 Control reference: Envelopes and pitch

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
Amp attack	0 to 20000 ms	0 ms
Amp decay	0 to 20000 ms	100 ms
Amp sustain	0 to 100 %	100 %
Amp release	0 to 30000 ms	100 ms
Pitch envelope	-48 to 48 st	0 st
Pitch attack	0 to 20000 ms	0 ms
Pitch decay	0 to 20000 ms	100 ms
Pitch sustain	0 to 100 %	0 %
Pitch release	0 to 30000 ms	100 ms
Fine tuning	-100 to 100 cents	0 cents
Velocity to volume	0 to 100 %	100 %
Velocity to pitch	-24 to 24 st	0 st
Random pan	0 to 100 %	0 %
Pitch bend range	0 to 24 st	0 st
Key label (manual override)	Auto detect, C major, C# major, D major, D# major, E major, F major, F# major, G major, G# major, A major, A# major, B major, C minor, C# minor, D minor, D# minor, E minor, F minor, F# minor, G minor, G# minor, A minor, A# minor, B minor	Auto detect

33 Control reference: Filter

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
Filter	Off, On	Off
Filter type	Low-pass, High-pass, Band-pass, Notch	Low-pass
Slope	12 dB/oct, 24 dB/oct	12 dB/oct
Cutoff	20 to 20000 Hz	20000 Hz
Resonance (Q)	0.1 to 10	0.707
Drive	0 to 24 dB	0 dB
Filter envelope	-8 to 8 oct	0 oct
Filter attack	0 to 20000 ms	0 ms
Filter decay	0 to 20000 ms	100 ms
Filter sustain	0 to 100 %	0 %
Filter release	0 to 30000 ms	100 ms
Velocity to filter	-8 to 8 oct	0 oct

34 Control reference: LFO

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
LFO	Off, On	Off
Waveform	Sine, Triangle, Square, Saw up, Saw down, Random	Sine
Rate	0.01 to 30 Hz	1 Hz
Clock	Free Hz, Host tempo	Free Hz
Beat division	1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1 bar, 2 bars, 4 bars	1/4
LFO fade in	0 to 10000 ms	0 ms
LFO retrigger	Free-running, Per note	Per note
Phase	0 to 360 deg	0 deg
Volume depth	0 to 100 %	0 %
Pitch depth	0 to 24 st	0 st
Pan depth	0 to 100 %	0 %
Filter depth	0 to 8 oct	0 oct

35 Control reference: Outputs and sends

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
Audio output	Main, Out 2, Out 3, Out 4, Out 5, Out 6, Out 7, Out 8, Out 9, Out 10, Out 11, Out 12, Out 13, Out 14, Out 15, Out 16	Main
Send A	-60 to 0 dB	-60 dB
Send B	-60 to 0 dB	-60 dB
Send C	-60 to 0 dB	-60 dB
Send D	-60 to 0 dB	-60 dB
Send E	-60 to 0 dB	-60 dB
Send F	-60 to 0 dB	-60 dB

36 Control reference: Attack layer

Values are from the 0.11.1 control registry. Defaults apply to a new instrument or a page reset. A macro mapping can override the stored base value.

Control	Range / choices	Default
Attack layer	Off, On	Off
Attack sound	Click, Pulse, Noise	Click
Attack level	-60 to 0 dB	-18 dB
Attack tone	30 to 16000 Hz	1800 Hz
Attack decay	1 to 100 ms	8 ms
Position	0 to 2000 ms	0 ms